

Indian Institute of Information Technology Vadodara - International Campus Diu



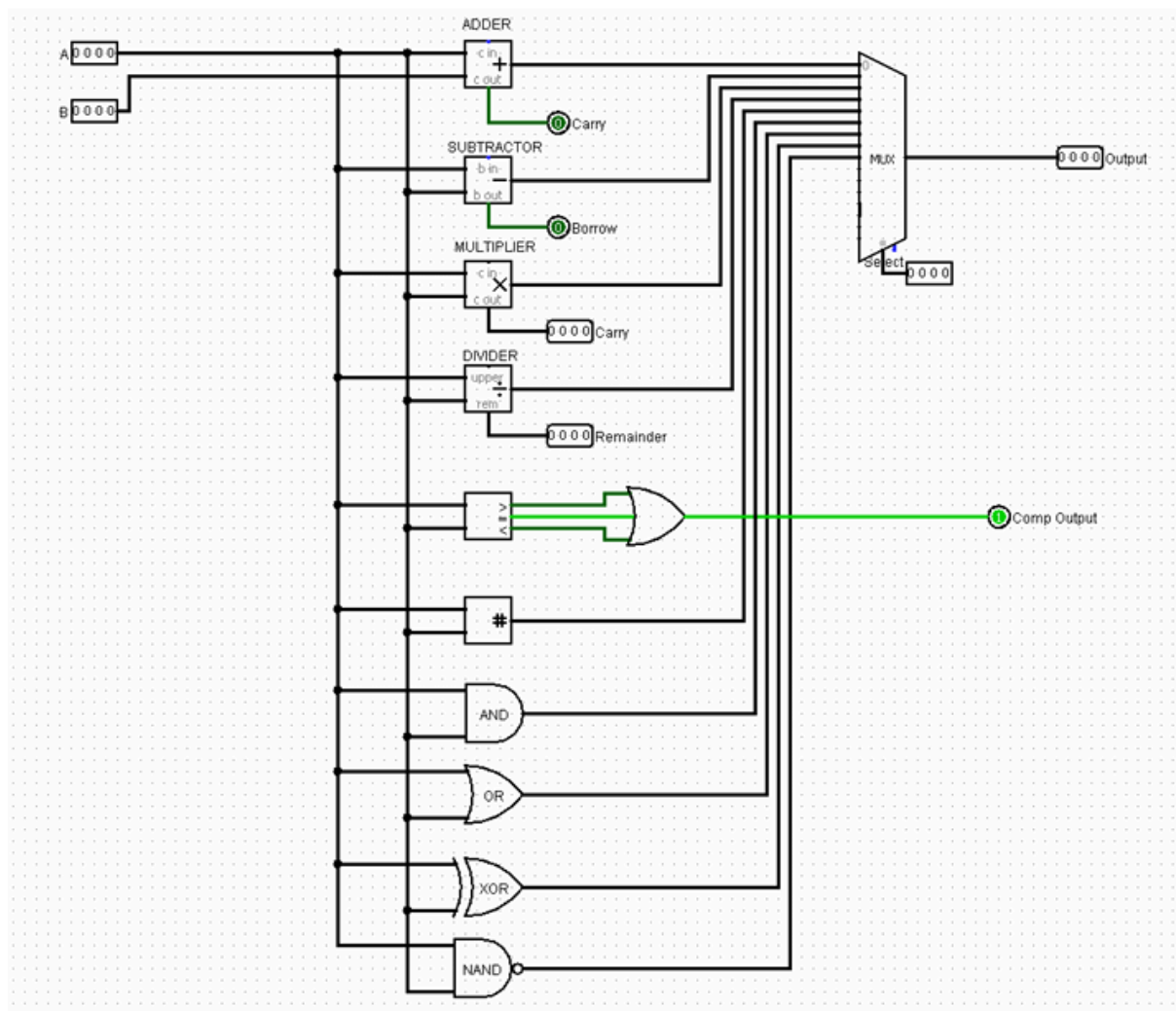
Name: **Saumya Joshi**
Roll no.: **202311039**
Branch: **CSE**
Batch: **2023**
Subject: **CS 268**

Aim:

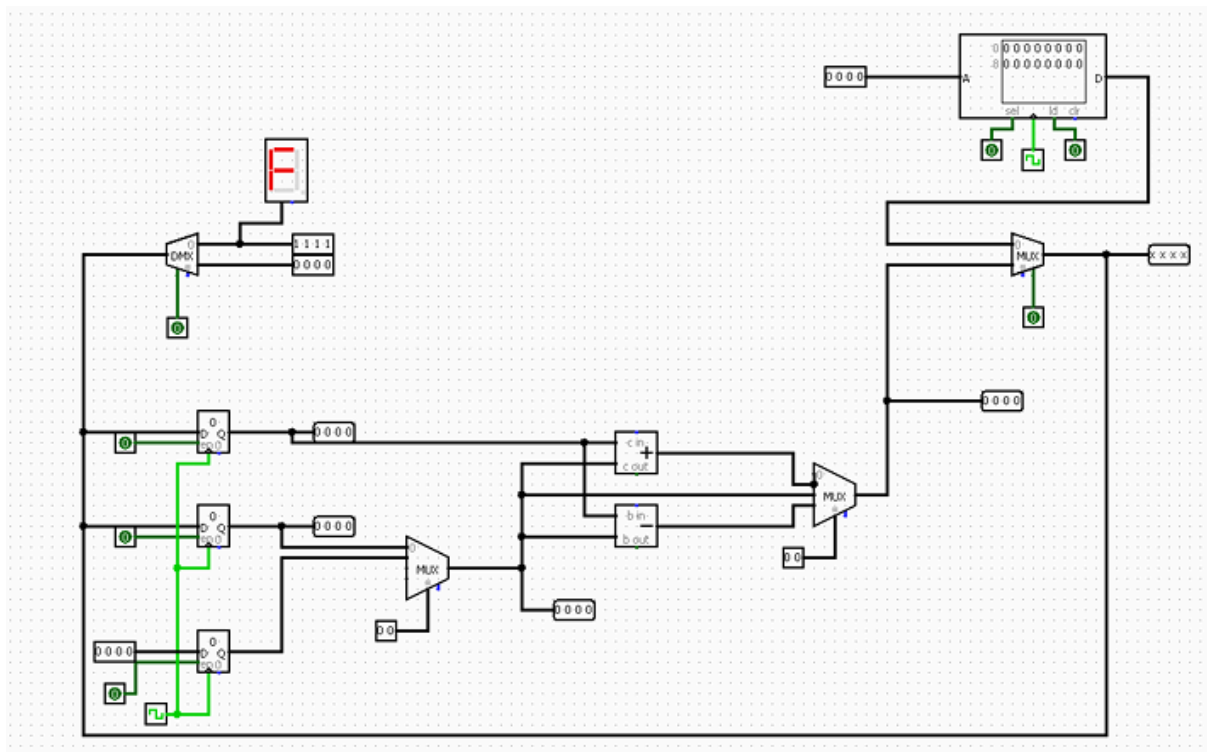
To design data path architecture based on a 4 Bit ALU

LOGISIM DIAGRAM:

4 Bit ALU



Data Path Architecture



Questions:

Question

Why is ALU necessary in MIPS?

Solution

ALU, Arithmetic Logical Unit is a central part of MIPS, a Microprocessor without Interlocked Stages. A microprocessor's main purpose is to process input data, but it cannot do that without an ALU. ALU forms the core computational logic required for the device to function.

Question

Explain Registers

Solution

Registers are high-speed memory units built directly into the CPU. They store data, addresses, and intermediate results during program execution. The reason for their use is that they are much faster than the main memory. They lead to an increase in efficiency due to the increased speed of accessing data.

Question

How read/write implementation happens in data path architecture?

Solution

In data path architecture, read/write operations involve generating and decoding addresses, transferring data via buses, and using control signals to coordinate the process. Data is fetched from memory or registers during a read and sent to the destination. During a write, data is stored at the specified address. The control unit manages these operations, ensuring synchronization with the clock cycle for efficient data processing and storage.

Question

Explain Data and instruction memory?

Solution

Instruction memory and data memory are critical components in computer architecture. Instruction memory stores the program's code, which the CPU fetches and executes, while data memory holds the variables, constants, and results that the program operates on. Instruction memory is typically read-only during execution and accessed via the program counter, whereas data memory is read-write and accessed using memory addresses.

Question

How many 16-bit ALU slices can be used for designing a 64-bit ALU?

Solution

$64/16 = 4$ slices of 16 Bit ALU's would be required to design a 64 bit ALU.

Result:

The data path architecture has been constructed.